

Pure-State Tangent Spectral Flow of a Frozen Adaptive-Circuit Record: CPU Pilot Protocol

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¹*Class-A fermionic adaptive-circuit project*
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We specify a CPU pilot that resolves the projective many-body singular gaps of a trajectory-resolved adaptive Chern circuit as boundary flux is threaded. For each of a soft explicit interface and a hard support-terminated wall, one zero-flux Born trajectory is generated with a raster- y measurement schedule and then replayed at 17 static twists in each orientation. The geometry is $N_x \times N_y = 16 \times 20$, the duration is 40 complete cycles, and all runs use a pure half-filled initial state, $n_{\text{shell}} = 1$, perfect correction, no postselection, and complex128 arithmetic. The Gaussian tangent cocycle is initialized in the exact occupied–empty decomposition of the cycle-zero Slater determinant and accumulated without a burn-in reset over the complete 40-cycle word. Its one-leg singular values generate the coherent one-particle–one-hole sector of the projectivized many-body singular spectral singular spectrum. We save up to 64 finite low-gap candidates, require at least 32, and continuously follow 32 branches by their occupied- and empty-leg overlaps. Wall-resolved weights define a spectral particle–hole excitation charge. This quantity is neither a fractional occupation of the evolving pure state nor a physical pumped charge, and no quantization criterion is imposed. This note fixes the protocol, estimators, continuation rule, numerical checks, output contract, and claim boundary. New circuit results are intentionally deferred; the completed equilibrium flattened-Hamiltonian pump is included only as calibration and motivation.

I. QUESTION AND CLAIM BOUNDARY

The class-A adaptive protocol measures localized overcomplete-Wannier (OW) occupations and applies outcome-dependent fermionic feedback without postselection [1]. For a fixed trajectory record $\mathbf{m}$, the ordered many-body evolution operator (mEO) is denoted $\hat{V}_{\mathbf{m}}$. We ask whether its slow projective tangent sector contains wall-localized modes whose spectral identity winds under a boundary twist ϕ .

This question differs from the equilibrium wall-charge pump. In the latter, one diagonalizes a static single-particle Hamiltonian, continuously transports its occupied projector around a flux loop, and compares the regional charge of that projector at the endpoints. In the circuit, the physical pure-state projector generated by independent static replays closes under a large gauge transformation. Continuing that projector therefore reproduces the static closure already tested by the preceding pilot. The present experiment instead continues elementary one-particle–one-hole tangent modes of the normalized trajectory map.

Accordingly, the pilot may establish a flux-dependent projective gap, wall localization, branch exchange, and a signed *spectral excitation charge*. It cannot by itself establish an integer Hall pump, a current integrated over monitored time, or a fractional occupation of a Lyapunov mode. Neither an integer excitation charge nor exact spectral closure is an acceptance gate.

II. COMPLETED EQUILIBRIUM CALIBRATION

The prior equilibrium calculation provides the target phenomenon but not a proof that the circuit tangent spectrum realizes it. For OW trial family $\nu = A, B$ and band $n = \pm$, it constructs

$$F_{\nu n}(\phi) = \sum_{\mathbf{R}} |W_{\mathbf{R}, \nu n}(\phi)\rangle \langle W_{\mathbf{R}, \nu n}(\phi)| \quad (1)$$

and the flattened one-body operator

$$H_{\text{flat}}(\phi) = F_{A+} + F_{B+} - F_{A-} - F_{B-}. \quad (2)$$

The benchmark used $N_x \times N_y = 20 \times 40$, 1600 one-particle orbitals, rank 800, $n_{\text{shell}} = 1$, $\alpha_{\text{top}} = 1$, $\alpha_{\text{triv}} = 30$, and walls at $x = 5, 15$. Both untruncated and support-terminated OW constructions were tested.

Two state prescriptions answer different questions. The instantaneous projector occupies the globally lowest 800 eigenstates at every ϕ ; it is the finite-size ground state and closes after 2π . The continued projector chooses its occupied modes once at $\phi_0 = \pm 10^{-7}$ and matches them between neighboring fluxes by maximum overlap. A Hungarian assignment enforces a one-to-one match, while an SVD polar rotation aligns clusters split by less than 10^{-9} . Occupations are never reselected. This produces a diabatic continuation through the exponentially small wall anticrossing while remaining adiabatic to the bulk.

For regional projectors R_L, R_R , the reported transferred charge is

$$q_{\text{pump}} = \frac{1}{2} [\Delta \text{tr}(R_R P) - \Delta \text{tr}(R_L P)]. \quad (3)$$

At 20×40 and $\phi_0 = +10^{-7}$, the instantaneous result is 1.49×10^{-8} (untruncated) or 2.52×10^{-9} (truncated),

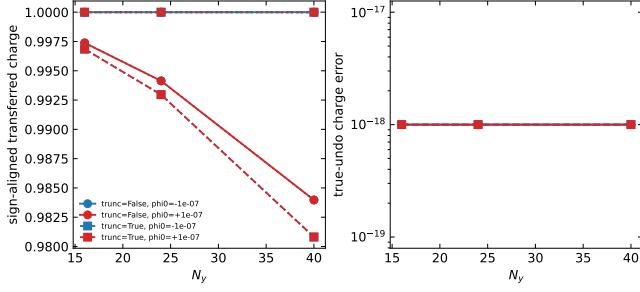


FIG. 1. Completed equilibrium flattened-Hamiltonian calibration. Left: sign-aligned continued wall-charge transfer for $N_y = 16, 24, 40$, both OW wall constructions, and both signs of the 10^{-7} starting offset. One winding is essentially integer; the partner reflects the initial opposite-wall weight. Right: a true reverse traversal, initialized from the forward endpoint, undoes the transfer to 10^{-18} -scale or better. These are previously completed equilibrium results, not outputs of the present circuit pilot.

whereas continued increasing flux gives 0.9839718573 and 0.9808096958. The opposite winding gives -0.9999999800 and -1.0000000032 . Charge conservation is satisfied within 1.14×10^{-13} and more than 0.9999987 of the transfer lies in a two-cell window around the physical walls. The noninteger 0.98 endpoint is stable on 129, 257, and endpoint-refined 751-point meshes: it reflects the regulated starting mode's small weight on the opposite wall, not inadequate flux resolution.

The true undo follows the cached forward path backward from its endpoint. For $N_y = 16, 24, 40$, both wall constructions and both offset signs obey $q_{\text{undo}} = -q_{\text{forward}}$ to stored precision, with normalized projector return error between 5.3×10^{-22} and 4.2×10^{-18} ; see Fig. 1. This validates the continuation algorithm. It does not make two independently initialized opposite windings literal time reverses.

The quoted endpoint and mesh data are stored in `notebooks/flattened_hamiltonian_analysis/outputs/exact_wall_charge_pump_refined_v1/run_20260901_002219/`; the size, offset, and true-undo audit and the plotted source are stored in `notebooks/flattened_hamiltonian_analysis/outputs/exact_wall_charge_pump_reversibility_v1/run_20260901_162444/`. Both directories contain machine-readable manifests and CSV tables in addition to the displayed figure.

The circuit pilot deliberately does not transplant Eq. (3) onto its closing physical projector. Instead it asks whether the low-gap tangent excitations display an analogous exchange of wall character. The equilibrium result therefore fixes the comparison language and expected spatial signature, while quantization remains outside the circuit acceptance criteria.

III. LOCKED CIRCUIT PROTOCOL

A. Geometry and dynamics

Both wall constructions use the parameters in Table I. A cycle is a raster- y ordered composition of OW-number measurements with conditional fermionic swaps. Distinct OW stabilizers need not commute, and the raster convention is therefore part of the scientific definition of the ordered word rather than a performance option.

The soft arm retains the OW tails throughout the cylinder and measures every site. The hard arm truncates the OW construction to the prescribed domain and performs the canonical exterior preparation before measuring only the active slab. Each arm receives an independently generated zero-twist reference because its active channel sequence and Born outcomes differ.

The earlier static-response campaign used random-series records. Those files are retained as immutable provenance but are not replayed here. Reordering a saved stochastic word would define a different noncommuting circuit. This pilot therefore generates new raster- y reference records in its own versioned result directory.

B. Frozen record and signed twist grid

For each arm, a single zero-twist trajectory produces a cycle-zero correlation matrix G_0 , the final G_{40} , and the ordered record $\mathbf{m}$. A forced replay at zero twist must reproduce G_{40} before any flux task is admitted. The same G_0 and the same outcomes, feedback targets, and ordered site list are then imposed at every twist in both orientations.

TABLE I. Locked numerical contract. The independent unit of computation is one static twist replay of one immutable wall-specific record.

quantity	value
geometry	$N_x \times N_y = 16 \times 20$
physical cycles	$40 = 2N_y$
OW support	$n_{\text{shell}} = 1$
mass parameters	$\alpha_1 = 1, \alpha_2 = 30$
domain interval	$x \in [4, 12]$
initial state	pure, half filled
measurement order	raster- y
feedback	perfect correction
postselection	disabled
arithmetic	complex128
twist points	17 per orientation
tangent window	cycles 1–40
saved candidates	up to 64 finite; at least 32
continued branches	32

For $\sigma = +1$ (counterclockwise) and $\sigma = -1$ (clockwise),

$$\phi_j^{(\sigma)} = -\sigma 10^{-7} + \sigma \frac{2\pi j}{16}, \quad j = 0, \dots, 16. \quad (4)$$

The two sweeps therefore share the same exactly zero-twist stochastic word but start on opposite sides of the potentially ambiguous crossing. The 10^{-7} regulator selects a reproducible side of that crossing while being negligible on the scale of the flux step. It can improve apparent branch quantization by preventing a numerical eigensolver from making an arbitrary choice inside a nearly degenerate subspace; it does not create topological quantization and is not included as evidence for it.

The cycle-zero state is placed in the uniform gauge

$$U_\phi = \exp(i\phi y/N_y), \quad G_0(\phi) = U_\phi G_0 U_\phi^\dagger, \quad (5)$$

and the OW modes are rebuilt at the same static ϕ before replaying $\mathbf{m}$ for 40 cycles.

IV. PURE-STATE TANGENT SPECTRUM

A. Projective one-particle-one-hole sector

Let $P_\star$ denote the cycle-zero occupied projector of the pure Slater determinant. Choose orthonormal occupied vectors $\{|v_i\rangle\}$ and empty vectors $\{|u_a\rangle\}$. A Hermitian tangent perturbation is generated by the coherent one-particle-one-hole direction

$$\delta G_{ai} = |u_a\rangle\langle v_i| + |v_i\rangle\langle u_a|. \quad (6)$$

For a trajectory-resolved Gaussian map, this Grassmannian tangent sector is the coherent one-particle-one-hole sector of the projectivized many-body singular spectrum [2, 3]. Exact gain, loss, and occupation projections may make the underlying transfer word singular; the normalized tangent map nevertheless remains well defined on its finite directions, with null directions recorded explicitly.

At the beginning of cycle one, the canonical CPU engine diagonalizes the pure projector and initializes complete occupied and empty one-leg frames. Every realized rank-one channel acts on their concatenation. At each completed cycle the two blocks are QR-stabilized independently, and their triangular factors are accumulated in separately scaled cores. No alignment window or accumulator reset is used: the resulting cocycle is the complete finite word from cycle 1 through cycle 40.

Let ℓ_i^{occ} and ℓ_a^{emp} be the logarithms of the final singular values of the oriented occupied and empty one-leg blocks. More explicitly, if $K_{\text{occ}} = KV_{\text{occ}}$ and $K_{\text{emp}} = KU_{\text{emp}}$ denote the restricted 40-cycle products, their stabilized singular decompositions are

$$K_{\text{occ}} = V_{\text{out}} e^{L_{\text{occ}}} V_{\text{in}}^\dagger, \quad (7)$$

$$K_{\text{emp}} = U_{\text{out}} e^{L_{\text{emp}}} U_{\text{in}}^\dagger. \quad (8)$$

Thus v_i is an output column of V_{out} and u_a is an output column of U_{out} ; the corresponding input columns are retained as well. Equivalently, these output columns diagonalize the two restricted $KK^\dagger$ blocks. They label coherent directions, not distinguishable particles in the reference trajectory. The repository convention forms the particle-hole tangent rate

$$\lambda_{ai} = \frac{\ell_i^{\text{occ}} + \ell_a^{\text{emp}}}{40}. \quad (9)$$

Equivalently, the empty-minus-occupied orientation is already absorbed into the definition of the occupied one-leg block. If the selected Slater ray is dominant, the corresponding projective excitation gap of

$$\hat{H}_{\text{eff}}(\phi) = -\log[\hat{V}_{\mathbf{m}}(\phi)\hat{V}_{\mathbf{m}}^\dagger(\phi)] \quad (10)$$

is

$$\Delta E_{ai} = -2(\ell_i^{\text{occ}} + \ell_a^{\text{emp}}), \quad \Delta \epsilon_{ai} = \frac{\Delta E_{ai}}{40} = -2\lambda_{ai}. \quad (11)$$

These are gaps relative to the selected projective ray. They do not determine the additive zero of the absolute many-body spectrum.

B. Candidate selection and spatial data

All finite pairs are constructed before truncation. Up to 64 candidates with smallest $|\lambda_{ai}|$ are saved at each twist, and the exact finite-pair and saved-candidate counts accompany every task. At least 32 are required because 32 branches are continued. For every candidate the output and input occupied and empty vectors, singular residual, pair rate, effective gap, cell density, x density, and wall weights are retained. Saving a pool twice as large as the reported branch set permits a continued mode to move away from the instantaneous 32 lowest gaps near an avoided crossing without being lost.

V. FLUX CONTINUATION AND WALL EXCITATION CHARGE

A. Common gauge and overlap assignment

Spatial probabilities are gauge invariant, but eigenvector overlaps between neighboring twists are not. Before continuation, every input and output mode is transformed back to the common zero-twist coordinate gauge by $U_\phi^\dagger$. Let (v_i, u_a) and (v'_j, u'_b) denote candidates at neighboring twists. Their matching score is

$$\mathcal{O}_{ai,bj} = |\langle v_i | v'_j \rangle| |\langle u_a | u'_b \rangle|. \quad (12)$$

A rectangular linear assignment maps the 32 continued branches into up to 64 finite current candidates by maximizing the total overlap. The run requires at least 32

finite candidates at every flux; exact null directions are never promoted into artificial modes merely to fill the table. A 10^{-6} -weighted gap distance breaks only numerical ties. Clockwise and counterclockwise branches are continued independently from their respective regulated origins. Step overlaps and endpoint closure overlaps are diagnostics and remain in the raw analysis product.

B. What “mode occupation” means

For a pure trajectory the state occupations of the two one-leg sectors are exactly one and zero. The nontrivial flux-dependent observables are their regional probability weights. With R_L and R_R the disjoint Voronoi basins of the walls at $x = 4$ and $x = 12$, define

$$p_{L/R}^{\text{occ}}(\phi) = \langle v_i(\phi) | R_{L/R} | v_i(\phi) \rangle, \quad (13)$$

$$p_{L/R}^{\text{emp}}(\phi) = \langle u_a(\phi) | R_{L/R} | u_a(\phi) \rangle. \quad (14)$$

The excitation projector obtained by removing v_i and adding u_a is

$$P_{ai} = P_\star - lvertv_i\rangle\langle v_i| + lvertu_a\rangle\langle u_a|. \quad (15)$$

Its normalized left–right excitation charge is

$$q_{ai}(\phi) = \frac{1}{2} [(p_R^{\text{emp}} - p_L^{\text{emp}}) - (p_R^{\text{occ}} - p_L^{\text{occ}})]. \quad (16)$$

Thus $q_{ai} \simeq +1$ places the added particle on the right wall and the hole on the left, while $q_{ai} \simeq -1$ reverses that assignment. A value near zero can mean same-wall support or delocalization; the four underlying weights are therefore always saved and must be inspected with q_{ai} .

The labels “occupied” and “empty” refer first to the cycle-zero sectors from which the tangent legs originate. To test their endpoint interpretation rather than assume it, the data product also stores $\langle v_i | P_T | v_i \rangle$, $\langle u_a | P_T | u_a \rangle$ and the projector residuals $\|P_T v_i - v_i\|$ and $\|P_T u_a\|$ after undoing the common flux gauge. These diagnostics state directly whether a reported output leg is occupied-like or empty-like in the final physical Slater determinant.

Equation (16) is a property of a derived spectral excitation. It is not the total charge transported by the monitored state. For comparison only, every replay also saves the final physical N_L , N_R , and N_{tot} used in the preceding static-response pilot.

VI. NUMERICAL VALIDATION AND DATA CONTRACT

The physical evolution always calls the canonical CPU entry point `classA_U1FGTN.run_markov_circuit`. Tangent tracking is an opt-in engine mode and does not duplicate or replace the Markov cycle loop. The CPU occupied–empty implementation is tested against the canonical GPU block convention on small pure systems and against a direct full-product Gram reconstruction.

Before production, the following checks are mandatory:

1. each raster- y reference has a valid result/completion pair;
2. a forced zero-twist replay reproduces the sampled final correlation matrix to normalized Frobenius error below 10^{-12} ;
3. all physical and tangent arrays are complex128 or float64 as appropriate;
4. charge continuity agrees with the integer feedback source to 10^{-9} ;
5. tangent branch probabilities are finite and no invalid branch is admitted;
6. selected singular-vector residuals remain below 10^{-8} ;
7. tangent observation leaves the physical final state unchanged on a common-seed regression.

There are $2 \times 2 \times 17 = 68$ independent twist tasks. Each writes one compressed NPZ followed by one completion JSON containing the task identity, configuration identity, reference hash, filename, byte count, and SHA-256. A restart skips a task only after both files verify. Missing, partial, or mismatched pairs are re-computed. The completed random-serial campaign remains untouched; the present products live under `results/N16x20_frozen_flux_pure_tangent_v1`.

The analysis product contains the continued 32-branch arrays, a flat CSV, and a JSON ledger of overlap and residual diagnostics. No acceptance field tests q_{ai} against an integer and no result is promoted to a pump invariant.

VII. EXECUTION

The production job uses 28 independent processes on physical CPU cores 28–55, with one BLAS thread per process. A single parent progress bar reports verified and completed twist points. The tmux session is named `frozen_flux_pure_tangent_N16x20_v1`; its persistent log and all scientific products reside in the versioned result directory. One hard and one soft task are first executed in a separate smoke directory. The branch analysis runs only after all 68 result/completion pairs verify.

VIII. DEFERRED INTERPRETATION

No numerical conclusion is part of this protocol document. A subsequent results amendment may discuss whether low-gap branches remain localized, exchange wall character, show orientation-dependent monodromy, or correlate with the static conditioned charge response. Such an amendment must retain the distinction among an equilibrium occupied-projector pump, a trajectory-conditioned static replay, a pure-state tangent excitation, and a physical time-dependent monitored pump.

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- [1] A. Bhuiyan, H. Pan, and C.-M. Jian, [Physical Review Research](#) **8**, 023147 (2026), [arXiv:2507.13437](#).
[2] A. Bhuiyan (2026), unpublished project note.
[3] A. Bhuiyan (2026), unpublished project note.